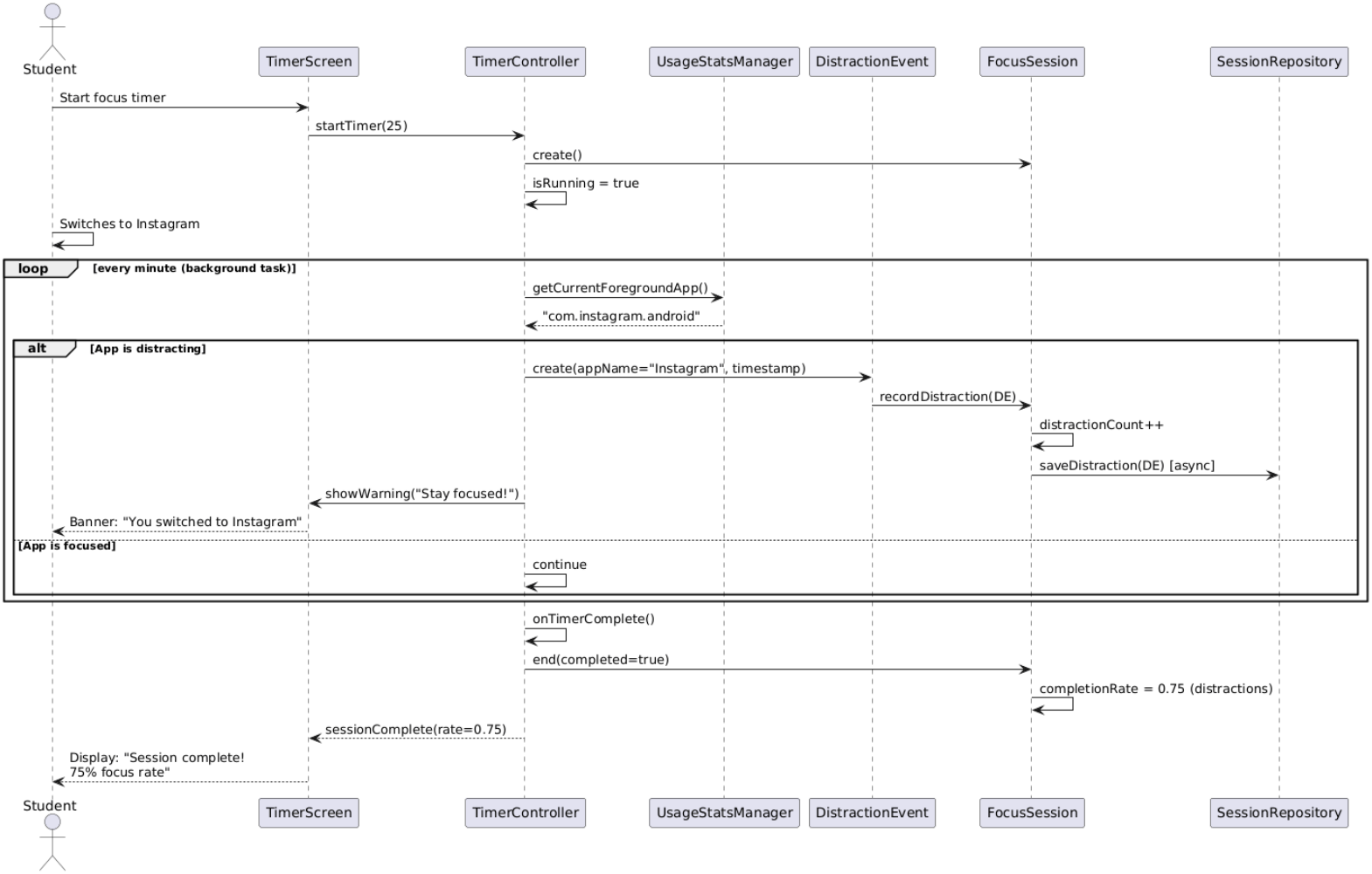
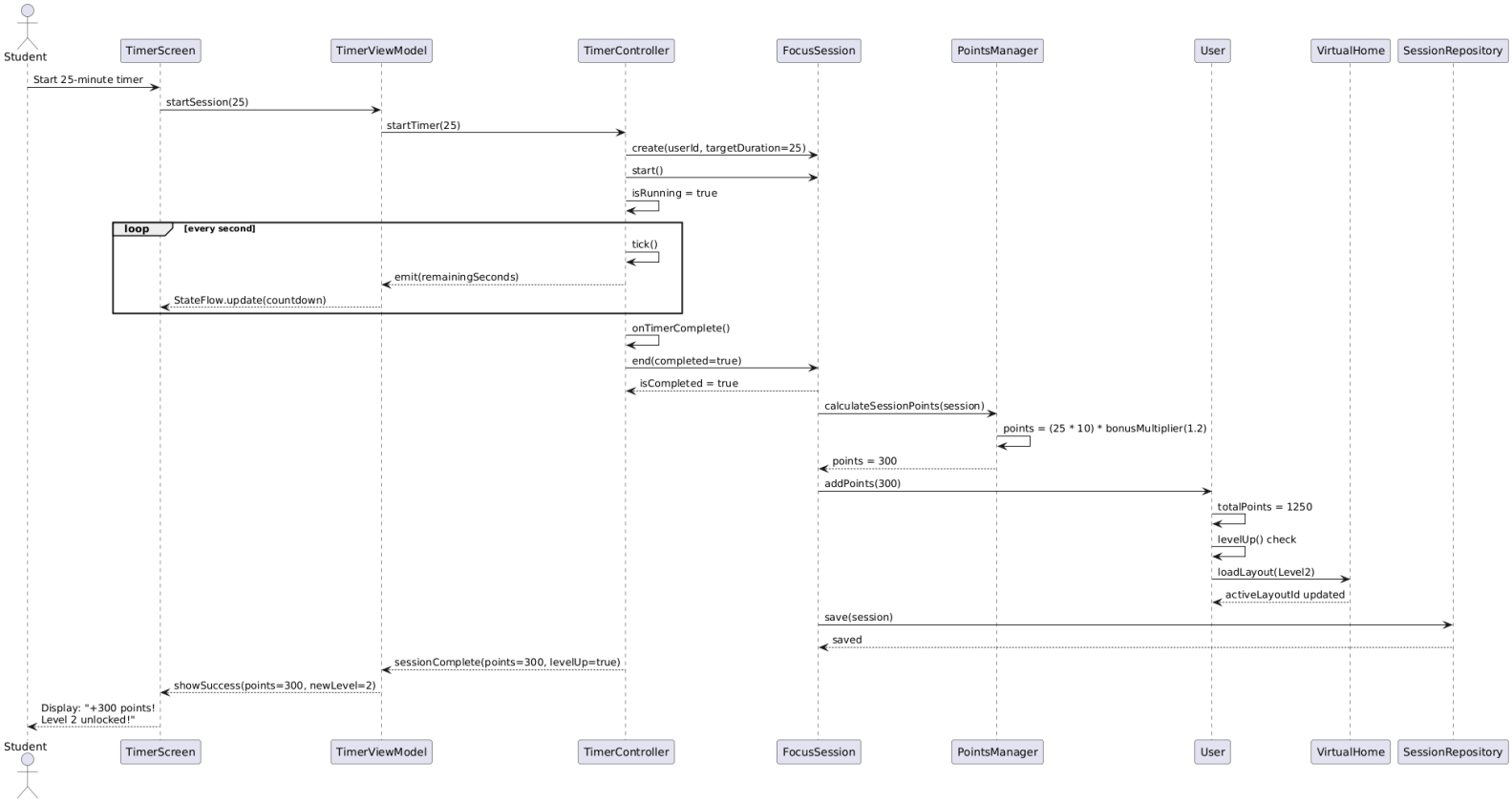
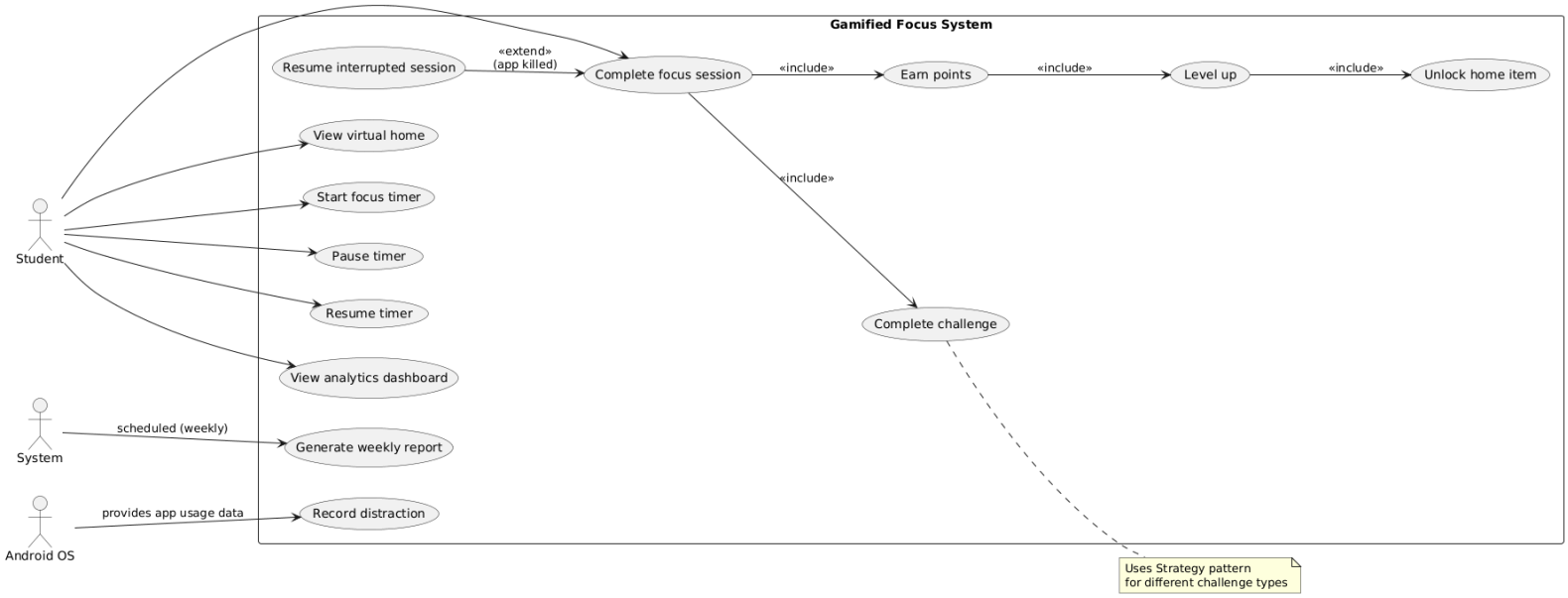
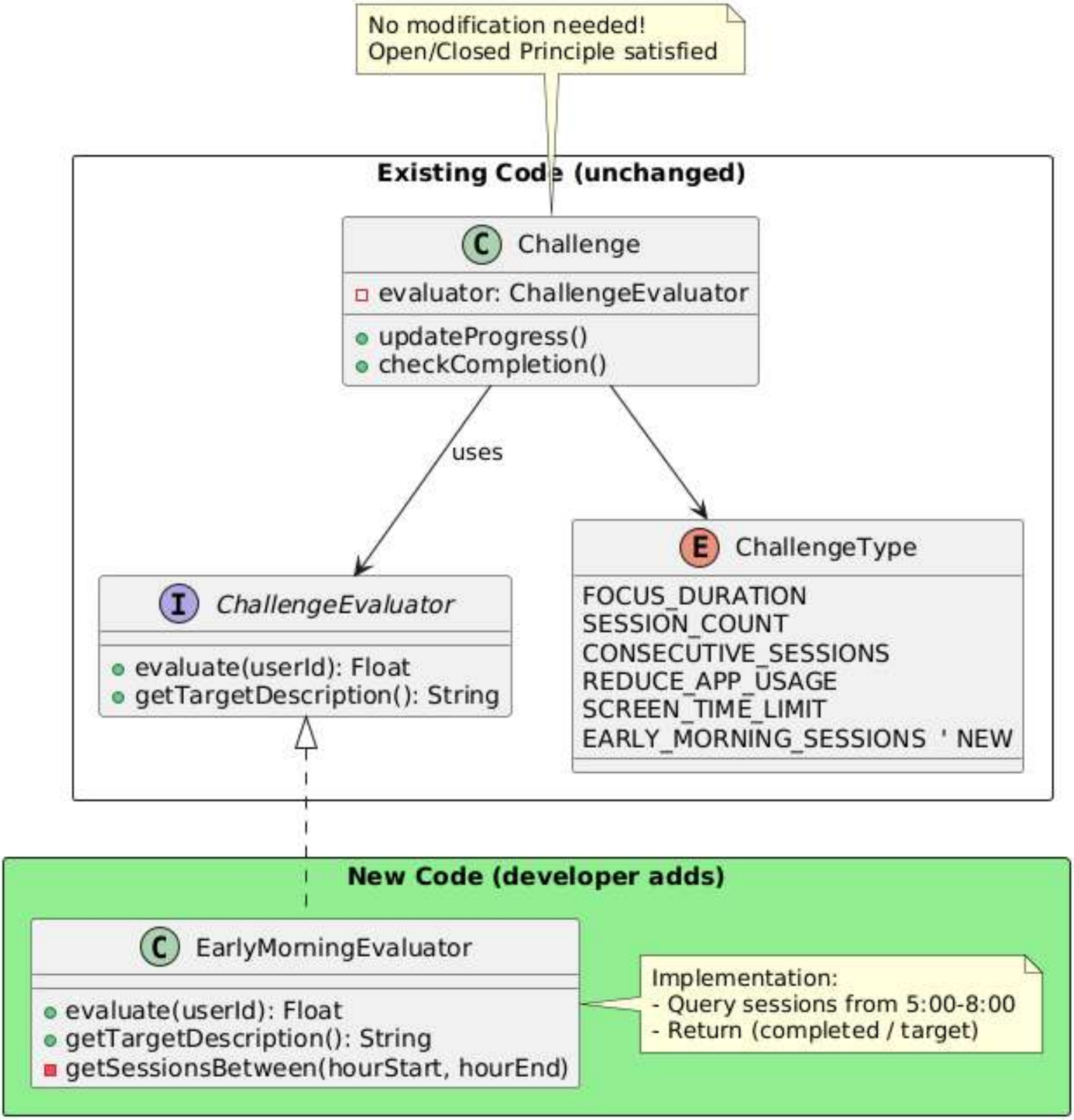
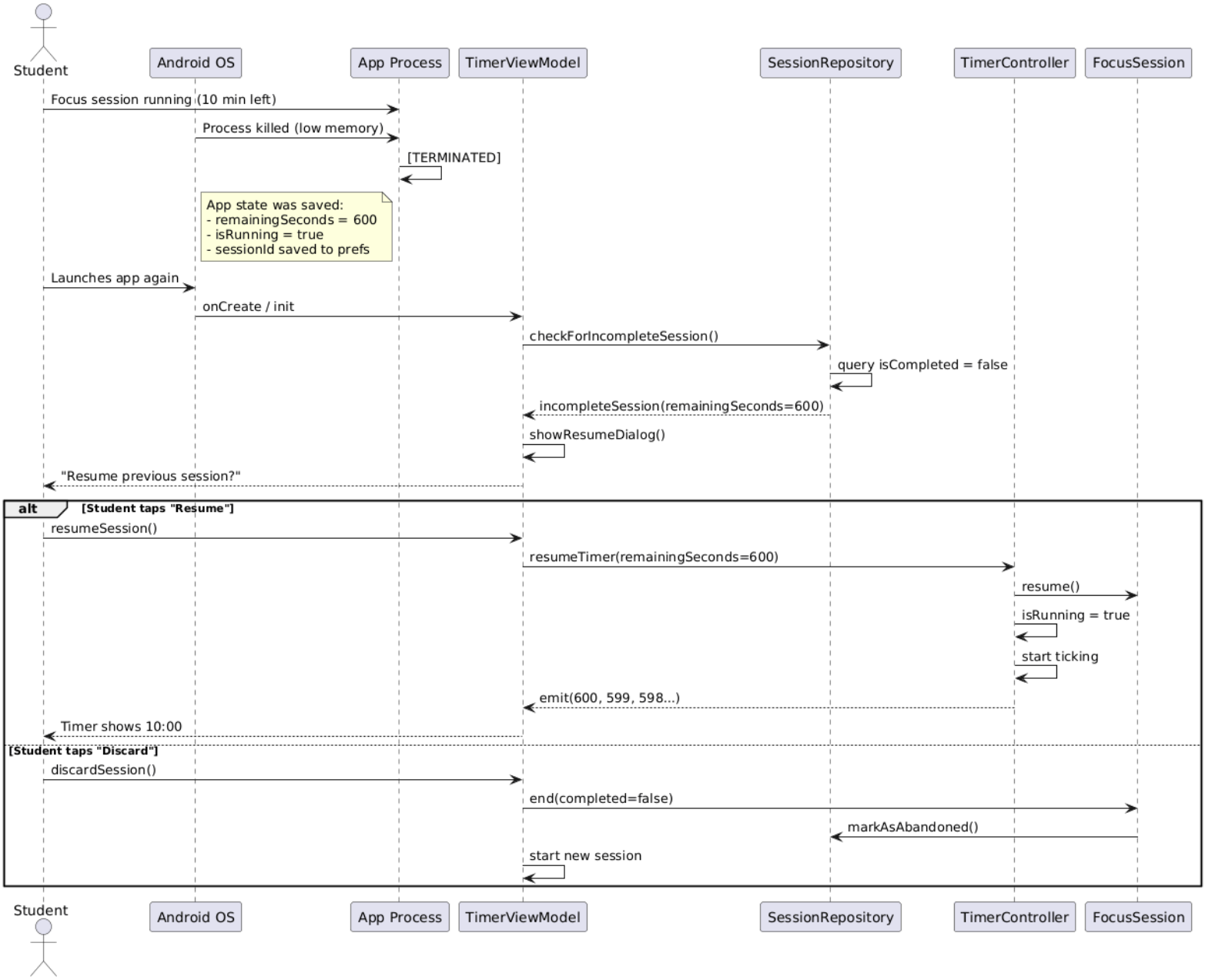
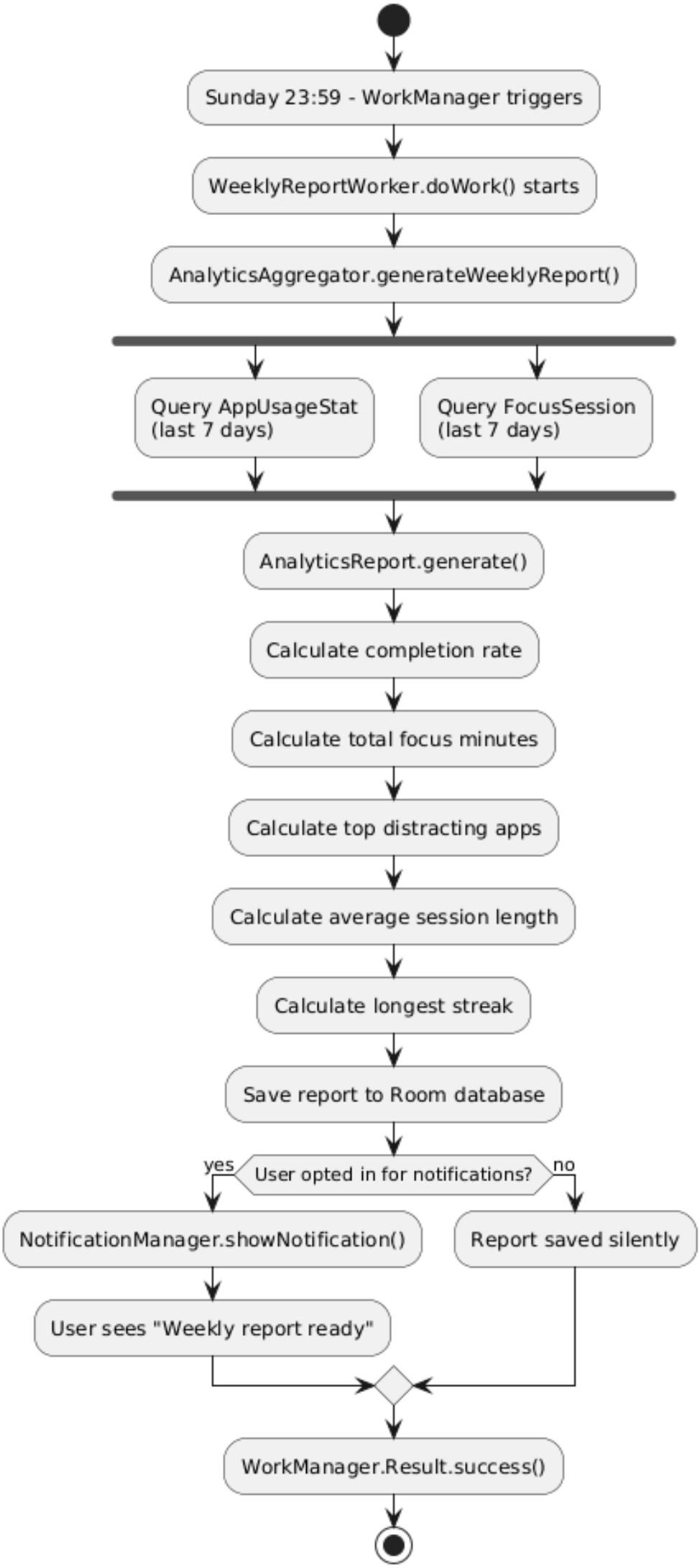

Student
TimerScreen
TimerController
UsageStatsManager
DistractionEvent
FocusSession
SessionRepository
Start focus timer
startTimer(25)
create()
isRunning = true
Switches to Instagram
loop
[every minute (background task)]
getCurrentForegroundApp()
"com.instagram.android"
alt
[App is distracting]
create(appName="Instagram", timestamp)
recordDistraction(DE)
distractionCount++
saveDistraction(DE) [async]
showWarning("Stay focused!")
Banner: "You switched to Instagram"
[App is focused]
continue
onTimerComplete()
end(completed=true)
completionRate = 0.75 (distractions)
sessionComplete(rate=0.75)
Display: "Session complete!
75% focus rate"

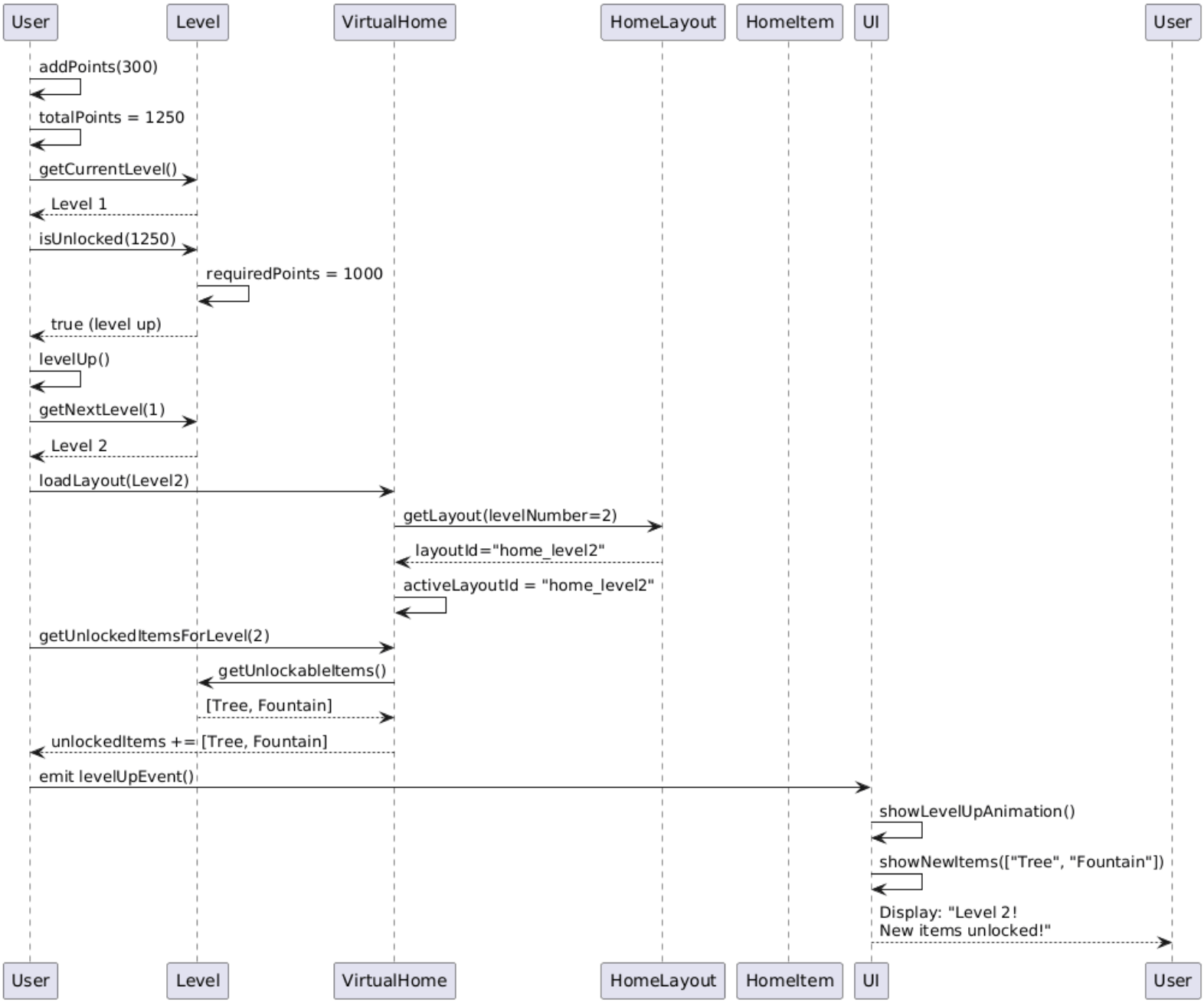

Student
TimerScreen
TimerViewModel
TimerController
FocusSession
PointsManager
User
VirtualHome
SessionRepository
Start 25-minute timer
startSession(25)
startTimer(25)
create(userId, targetDuration=25)
start()
isRunning = true
loop [every second]
tick()
emit(remainingSeconds)
StateFlow.update(countdown)
onTimerComplete()
end(completed=true)
isCompleted = true
calculateSessionPoints(session)
points = (25 * 10) * bonusMultiplier(1.2)
points = 300
addPoints(300)
totalPoints = 1250
levelUp() check
loadLayout(Level2)
activeLayoutId updated
save(session)
saved
sessionComplete(points=300, levelUp=true)
showSuccess(points=300, newLevel=2)
Display: "+300 points!
Level 2 unlocked!"

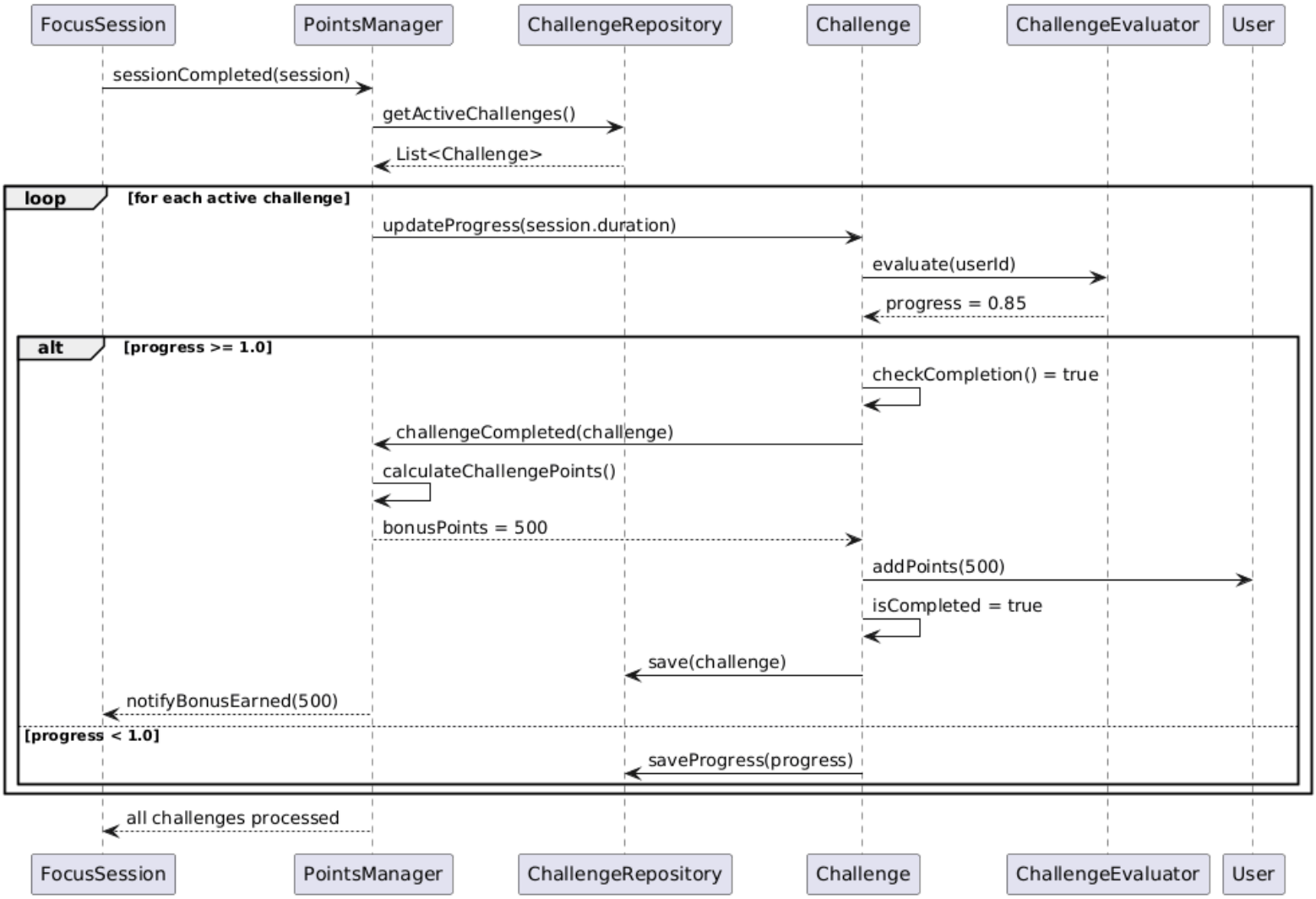

Gamified Focus System
Resume interrupted session
«extend»
(app killed)
Complete focus session
«include»
Earn points
«include»
Level up
«include»
Unlock home item
View virtual home
Start focus timer
Pause timer
Resume timer
View analytics dashboard
«include»
Complete challenge
Generate weekly report
Record distraction
Student
System
Android OS
scheduled (weekly)
provides app usage data
Uses Strategy pattern
for different challenge types


No modification needed!
Open/Closed Principle satisfied
Existing Code (unchanged)
Challenge
evaluator: ChallengeEvaluator
updateProgress()
checkCompletion()
uses
ChallengeEvaluator
evaluate(userId): Float
getTargetDescription(): String
ChallengeType
FOCUS_DURATION
SESSION_COUNT
CONSECUTIVE_SESSIONS
REDUCE_APP_USAGE
SCREEN_TIME_LIMIT
EARLY_MORNING_SESSIONS ' NEW
New Code (developer adds)
EarlyMorningEvaluator
evaluate(userId): Float
getTargetDescription(): String
getSessionsBetween(hourStart, hourEnd)
Implementation:
- Query sessions from 5:00-8:00
- Return (completed / target)


Student
Android OS
App Process
TimerViewModel
SessionRepository
TimerController
FocusSession
Focus session running (10 min left)
Process killed (low memory)
[TERMINATED]
App state was saved:
- remainingSeconds = 600
- isRunning = true
- sessionId saved to prefs
Launches app again
onCreate / init
checkForIncompleteSession()
query isCompleted = false
incompleteSession(remainingSeconds=600)
showResumeDialog()
"Resume previous session?"
alt
[Student taps "Resume"]
resumeSession()
resumeTimer(remainingSeconds=600)
resume()
isRunning = true
start ticking
emit(600, 599, 598...)
Timer shows 10:00
[Student taps "Discard"]
discardSession()
end(completed=false)
markAsAbandoned()
start new session


Sunday 23:59 - WorkManager triggers
WeeklyReportWorker.doWork() starts
AnalyticsAggregator.generateWeeklyReport()
Query AppUsageStat (last 7 days)
Query FocusSession (last 7 days)
AnalyticsReport.generate()
Calculate completion rate
Calculate total focus minutes
Calculate top distracting apps
Calculate average session length
Calculate longest streak
Save report to Room database
User opted in for notifications?
yes
no
NotificationManager.showNotification()
Report saved silently
User sees "Weekly report ready"
WorkManager.Result.success()


User
Level
VirtualHome
HomeLayout
HomeItem
UI
User
addPoints(300)
totalPoints = 1250
getCurrentLevel()
Level 1
isUnlocked(1250)
requiredPoints = 1000
true (level up)
levelUp()
getNextLevel(1)
Level 2
loadLayout(Level2)
getLayout(levelNumber=2)
layoutId="home_level2"
activeLayoutId = "home_level2"
getUnlockedItemsForLevel(2)
getUnlockableItems()
[Tree, Fountain]
unlockedItems += [Tree, Fountain]
emit levelUpEvent()
showLevelUpAnimation()
showNewItems(["Tree", "Fountain"])
Display: "Level 2!
New items unlocked!"
User
Level
VirtualHome
HomeLayout
HomeItem
UI
User


FocusSession
PointsManager
ChallengeRepository
Challenge
ChallengeEvaluator
User
sessionCompleted(session)
getActiveChallenges()
List<Challenge>
loop
[for each active challenge]
updateProgress(session.duration)
evaluate(userId)
progress = 0.85
alt
[progress >= 1.0]
checkCompletion() = true
challengeCompleted(challenge)
calculateChallengePoints()
bonusPoints = 500
addPoints(500)
isCompleted = true
save(challenge)
notifyBonusEarned(500)
[progress < 1.0]
saveProgress(progress)
all challenges processed
FocusSession
PointsManager
ChallengeRepository
Challenge
ChallengeEvaluator
User